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1 Implementation Plan: shared-fault cross-rank DOFData consistency (SAFS)

1.1 TL;DR – there are TWO separate problems; this plan is about the second one

	Problem A — the blow-up	Problem B — the cross-rank inconsistency
What	The fault +/- side was mislabeled on the tilted (curvilinear) fault, so the nucleation pushed with the wrong sign → V_max → 10 ³ blow-up.	The same fault quad-point is solved twice (once per MPI rank) from inputs that differ at the 16th digit; at the rupture onset that hair's-width difference makes the two ranks disagree (slip vs lock).
Scope	Geometry (which side is “+”). Independent of friction law / dip / strike.	MPI / floating-point. Triggered the same way for any channel; only runs away when there is dip slip.
Status	FIXED — commit 800b3281 (see Part A). Validated by the zerodip job.	OPEN — this plan.
Test	spatial_dyn_zerodip_8N_400r_dNew2localinjectiontest + the (no dip slip, guard non-fatal).	Dc2 XRANK Frontera run.

Documents behind this plan: diagnosis spatial_dynamic_rupture_speckle_blowup_2026-05-22.md, design review spatial_dynamic_rupture_reconcile_review_2026-05-23.md.

1.2 Part A – the blow-up fix (DONE, commit 800b3281) – documented for the record

This part is already landed and is NOT implemented by this plan. It is documented here because the user asked to keep it separate and because it is the reason the zerodip run is now stable.

What was wrong (plain language)

On each fault face the code must decide which of the two touching elements is on the “+” side and which is on the “–” side (this sets the sign of the nucleation push and the flux). It decided by measuring which element sits “below” a FIXED reference direction $\text{ref_normal} = (0, -1, 0)$ — i.e. it projected the element-to-face offset onto that hardcoded direction. That works for a flat fault whose normal IS $(0, \pm 1, 0)$. The meshed SAFS fault is **curvilinear**: where the fault swings to a roughly N–S strike, its normal becomes nearly perpendicular to $(0, -1, 0)$, so the projection measures mostly the *sideways* part of the offset instead of the real “above/below.” Past a tilt of $\approx 71.6^\circ$ the sideways part wins and the side label **flips** — both ranks of a shared face then call themselves “–”, the “exactly one + side” rule breaks, the nucleation is applied with the wrong sign, and the rupture blows up.

The exact computation (old vs new)

The side label is set in `wave_operator.inl` (interior :459–465, shared :519–525):

```
elem1_proj = elem1_c · ref_normal      face_proj = face_c · ref_normal
elem1_on_plus = (elem1_proj < face_proj) ⇔ (elem1_c — face_c) · ref_normal < 0
```

Let $\mathbf{d} = \mathbf{elem1_c} - \mathbf{face_c}$ be the vector from the face centroid to the owning element’s centroid. Split it into the part along the TRUE face normal $\hat{\mathbf{n}}$ and the in-plane (tangential) part: $\mathbf{d} = \mathbf{d_n} \cdot \hat{\mathbf{n}} + \mathbf{d_t}$, where by definition $\mathbf{d_t} \cdot \hat{\mathbf{n}} = 0$. The OLD test computes $\mathbf{d} \cdot \text{ref_normal}$:

$$\mathbf{d} \cdot \text{ref_normal} = \mathbf{d_n} (\hat{\mathbf{n}} \cdot \text{ref_normal}) + (\mathbf{d_t} \cdot \text{ref_normal})$$

- **Flat fault:** $\hat{\mathbf{n}} = \pm(0, 1, 0) = \pm \text{ref_normal}$, so $\hat{\mathbf{n}} \cdot \text{ref_normal} = \pm 1$ and the tangential term $\mathbf{d_t} \cdot \text{ref_normal} = 0$ (because $\mathbf{d_t} \perp \text{ref_normal}$ too). Result: $\mathbf{d} \cdot \text{ref_normal} = \pm \mathbf{d_n}$ — it measures exactly how far the element sits above/below the face. **Correct.**
- **Tilted fault (face normal makes angle θ with ref_normal):** $\hat{\mathbf{n}} \cdot \text{ref_normal} = \cos \theta$, and the tangential part now projects with weight $\sim \sin \theta$. A tetrahedron’s centroid is NOT directly over its face centroid, so $\mathbf{d_t} \neq 0$. Result: $\mathbf{d} \cdot \text{ref_normal} \approx \mathbf{d_n} \cos \theta + (\mathbf{d_t} \cdot \text{ref_normal})$. As $\theta \rightarrow 90^\circ$ (N–S strike, $\hat{\mathbf{n}} \perp \text{ref_normal}$), $\cos \theta \rightarrow 0$ and the **tangential** term takes over. When $|\mathbf{d_t} \cdot \text{ref_normal}| > |\mathbf{d_n} \cos \theta|$, the SIGN is set by the sideways offset, not the real above/below $\rightarrow$ **wrong label.**
- On the 2-tet test fixture this margin is $\mathbf{d} \cdot \text{ref_normal} = \text{kl} \cdot (\sin \theta/12 - \cos \theta/4)$, which crosses zero at $\tan \theta = 3 \rightarrow \theta = 71.6^\circ$: past that tilt the label flips, both ranks of the shared face get the same sign on $(\mathbf{elem1_c} - \mathbf{face_c}) \cdot \text{ref_normal} \rightarrow$ both call themselves “–” $\rightarrow$ “exactly one +” breaks $\rightarrow$ the fixed-sign nucleation vector is added to the wrong side $\rightarrow$ blow-up.

What we changed (commit 800b3281)

Project onto the ACTUAL face normal $\hat{\mathbf{n}}$ (canonicalized so both ranks agree on its direction), not the hardcoded `ref_normal`:

```
CalcOrtho(face Jacobian)  $\rightarrow$  n_raw
```

```
if NormalNeedsFlipToCanonical(n_raw, ref_normal, dim): n_raw = -n_raw // →  $\hat{n}$ 
elem1_on_plus = ((elem1_c - face_c) ·  $\hat{n}$  < 0)
```

Because $d_t \perp \hat{n}$ BY CONSTRUCTION, $d \cdot \hat{n} = d_n$ **exactly** — the tangential offset can never contaminate the sign, at any tilt. The decision margin is now $\pm |d_n| \sim 0(\text{element size})$, independent of θ , so it never flips. - `fault/fault_basis.hpp` — new shared static `FaultBasis::NormalNeedsFlipToCanonical(n_raw, ref_normal, dim)`: orient the normal LINE so $\hat{n} \cdot \text{ref_normal} > 0$; in the degenerate band where $\hat{n} \perp \text{ref_normal}$ (the $\theta=90^\circ$ case where even that sign is ill-defined) fall back to “largest- magnitude component positive.” Both ranks then compute the SAME $\hat{n}$ regardless of which way `CalcOrtho` happened to point it. `ComputeOrientedFrame` routes its sign decision through the same helper. - `dynamic/wave_operator.inl` — both side blocks compute $\hat{n}$ via `CalcOrtho` and project the centroid offset onto it instead of `ref_normal`. - For a flat fault $\hat{n} = \pm \text{ref_normal}$ so the boolean is identical to before (TPV/BP5 byte-exact); the change only differs where $\theta > 0$.

How Part A is validated (per the user’s directive)

Job script: `jobs/safs/spatial_dyn_zerodip_8N_400r_dev_2hr_safs.sbatch`. - `SEAS_ZERO_DIP_PRESTRESS=1` (default) — zeroes the dip pre-stress at the source so there is **no dip slip**; the fault is loaded in strike only. - `SEAS_R101_NONFATAL=1` — the cross-rank guard logs `worst_rel` each check instead of aborting, so **Problem B does not force-abort the Problem-A run** (this is the “do not force guard the MPI DOFData inconsistency” requirement; the mechanism already exists, commit `fc28454`, and the call site is correct: `spatial_dyn_driver.cpp:1994` passes `tol`, `&worst_rel`, `&worst_field`, `abort_on_fail!=!nonfatal`). - Result (job 7747119, MEASURED): the run advances with **no V_max blow-up** ($V_{\max}$ peaks ~ 6.3 m/s then decreases) — **Part A is fixed**. Crucially, the R-101 non-fatal log still goes `worst_rel` → `1.0` at $t \approx 0.595$ s (in the STRIKE fields `V2/slip2`): so Part A removed the *blow-up*, but **issue B (the cross-rank speckle) is still present** — direction-blind, just non-catastrophic without dip.

Acceptance for Part A (met): $V_{\max}$ physical (no 10^3 , peaks ~ 6 then decays). The persistent `worst_rel` → `1.0` is **issue B** (confirmed active); the speckled slip-rate elements coincide with it *in time* (working hypothesis: the speckle is issue B, to be confirmed/refuted by the Phase-4 post-fix rerun) — either way NOT a Part-A regression.

1.3 Part B – the cross-rank DOFData inconsistency (THIS PLAN)

B.1 The problem in plain language

A fault quadrature point that lies on the boundary between two MPI ranks is a **shared** point: rank A owns the element on one side, rank B owns the element on the other. To advance the fault friction there, each rank needs the stress on **both** sides. Each rank already has its own side exactly, and gets the other side as a **ghost copy** from its neighbor. So both ranks have “both sides” and both run the friction solve for that point. **The same physics is computed twice, once on each rank.** (This is what R-701 set up: it deleted an older “one rank computes, tells the other” broadcast, betting that both ranks would get identical answers anyway.)

That bet is wrong by a hair. The two ranks do **not** feed the friction solve bit-identical numbers. The reason is pure floating-point bookkeeping, not a logic bug:

- The shared point is one physical location, but rank A reaches it through its element's local coordinates and rank B reaches the *same* point through the *neighbor* element's local coordinates. The interpolation weights (shape × DOFs) are mathematically equal but rounded differently, so the bulk stress the two ranks plug in differs at about the **16th significant digit** (~1e-14). (Confirmed on Frontera: the two ranks' inputs match to 11 printed digits and differ below that.)

Normally a 1e-14 difference in → a 1e-14 difference out: invisible. The trouble is the **rupture onset is a yes/no switch**. The slip rate is $V = \max(0, (|\tau| - \text{strength})/\eta_s)$: below the strength the point is **locked** ($V=0$); above it, it **slips**. Right at the threshold $|\tau| = \text{strength}$, a hair's-width difference in $|\tau|$ flips the answer between “locked” and “slips.” The SAFS nucleation ramps the stress up **slowly**, so the point sits **right on the knife's edge for several sub-steps** — and during that window the 1e-14 difference makes **rank A slip while rank B stays locked** (or vice-versa). From that instant the two ranks are computing two different earthquakes at that point; the slip-weakening feedback then drives them apart, and the run is corrupted.

A runtime guard (the R-101 check) compares the two ranks each step and aborts when they disagree — that is what stops the SAFS run at $t \approx 0.455$ s.

Why benchmarks never caught it: TPV102/104 use rate-and-state friction, whose slip rate is a **smooth** function (no yes/no switch) — a 1e-14 input stays a 1e-14 output forever, so the ranks never visibly disagree. TPV205 uses the same slip-weakening switch as SAFS, but its rupture front sweeps past a point in one step (no slow knife's-edge dwell). Only SAFS combines the slip-weakening switch **and** a slow nucleation that parks a shared point on the threshold.

B.1.5 Root-cause status – what is **CONFIRMED** vs what is still **INFERRED**

Be honest about the chain, because it decides which fix is correct.

Confirmed (Frontera XRANK trace, job 7747036): - The shared-QP friction state is computed **redundantly** on both ranks. - At the diverging QP the two ranks' inputs match to **11 printed digits** yet the friction outputs **split** (one slips, one locks) at the $\max(0, \cdot)$ kink. Since the solve is deterministic, identical-to-11-digits inputs producing different outputs **proves there is a sub-11-digit cross-rank input difference**, and that the non-smooth onset amplifies it. This is the *mechanism* of the divergence.

Was open — now MEASURED and RESOLVED (Phase 0, job 7747304); see the RESOLVED block below. The three candidate sources were: - The **magnitude** of that input difference (“~1e-14” is an assumption, never measured). - Its **source**, which is one of: 1. **Interpolation:** the same physical QP is evaluated through two different element parametrisations (shape1•Loc1 on rank A vs shape2•Loc2 on rank B's ghost) — mathematically equal, differently rounded. Inherent to DG shared faces; unavoidable. 2. **Ghost-exchange not bit-exact:** rank B's ghost copy of rank A's element stress should be bit-identical to rank A's own, but isn't. A real defect — potentially fixable AT THE SOURCE. 3. **Accumulated bulk divergence:** the distributed bulk solve drifts across ranks over the run (different reduction/matvec order). Generally unavoidable.

Why this matters for the fix (decision gate): - Source = (1) or (3) → the gap is inherent → the **reconcile** (Phase 2) is the correct, general fix. - Source = (2) → make the ghost copy bit-exact → both ranks get identical inputs → identical outputs → **no reconcile needed** (this would vindicate

R-701's no-broadcast design and be cheaper/more local).

RESOLVED — Phase 0 ran (oblique non-fatal job 7747304, 2026-05-23): source (1) interpolation. At the diverging QP (607517.9, 3706359.2, -4543), shared by ranks 104 (owns +) and 114 (ghosts +), the $\%.17e + \text{RAWDof}$ trace shows: - the **raw +side element DOFs are bit-identical** across the two ranks at every common sub-step, in all of SXX/SXY/SXZ → **rules out source (2)** (ghost-exchange is bit-exact) **and source (3)** (no drift; identical all the way to $t \approx 1.52$ s); - the **interpolated** canonical-frame inputs differ at $\sim 1e-14$ (e.g. +side dip shear $4.27409779095866838e+02$ vs $\dots 725e+02$; -side $\dots 159172e-01$ vs $\dots 159116e-01$) → the seed is the unavoidable DG shared-face interpolation gap (shape1•Loc1 vs shape2•Loc2); - at the slip-onset kink that $1e-14$ tips the ranks onto opposite branches: $V1 = -1.637$ m/s (rank 104, slipping) vs 0 (rank 114, locked) → $\text{worst_rel}=1$ at $t \approx 0.49$ s, field $V1$ (dip), spreading to slip2.

⇒ **The reconcile (Phase 2) is the correct, necessary fix; a source-fix is NOT available** (the DOFs are already bit-exact — the gap is in interpolation). R-701's "both ranks agree anyway" premise is disproved. The oblique run is **bounded** (V_{max} peaks ~ 6 m/s; dip slip does NOT run away after Part A) and reproduces the speckle ($\text{max_slip} \rightarrow 1.27e6$ m, unphysical, direction-blind — same as $\text{zerodip } 4.45e6$ m) as the Phase-4 final-check baseline.

B.2 Why the DIP direction is the one flagged – what the equations actually say

I earlier hand-waved "dip couples to normal stress." You asked for equations, and they are worth being exact about, because **the local fault solve is dip/strike SYMMETRIC** — there is no built-in per-direction instability, and any claim that "dip is unstable, strike is stable" is wrong at the solve level.

The fault-local solve. In the canonical (fault-aligned) frame the bulk state splits into THREE independent Riemann channels — normal (local X), dip ($Y=t1$), strike ($Z=t2$) — `ComputeTrialTraction`, `fault_face_flux.cpp:49`:

$$\sigma_{\text{n_trial}} = \eta_{\text{p}} (v_{\text{n}^-} - v_{\text{n}^+} + \sigma_{\text{n}^+}/Z_{\text{p}^+} + \sigma_{\text{n}^-}/Z_{\text{p}^-}) \quad (\text{normal, Eq.7a})$$

$$\tau_{1\text{_trial}} = \eta_{\text{s}} (v_{\text{t1}^-} - v_{\text{t1}^+} + \tau_{1^+}/Z_{\text{s}^+} + \tau_{1^-}/Z_{\text{s}^-}) \quad (\text{dip, Eq.7b})$$

$$\tau_{2\text{_trial}} = \eta_{\text{s}} (v_{\text{t2}^-} - v_{\text{t2}^+} + \tau_{2^+}/Z_{\text{s}^+} + \tau_{2^-}/Z_{\text{s}^-}) \quad (\text{strike, Eq.7c})$$

Friction (`SolveLSW_TPV205`, `tpv205_friction.hpp:146-165`):

$$|\tau| = \sqrt{\tau_{1\text{_total}}^2 + \tau_{2\text{_total}}^2} \quad \tau_{\text{str}} = \mu_{\text{eff}} \cdot |\sigma_{\text{n_total}}|$$

$$V_{\text{abs}} = \max(0, (|\tau| - \tau_{\text{str}})/\eta_{\text{s}})$$

$$V1 = V_{\text{abs}} \cdot \tau_{1\text{_total}}/|\tau| \quad V2 = V_{\text{abs}} \cdot \tau_{2\text{_total}}/|\tau|$$

$$\tau_{1\text{_corr}} = \tau_{1\text{_trial}} - \eta_{\text{s}} \cdot V1 \quad \tau_{2\text{_corr}} = \tau_{2\text{_trial}} - \eta_{\text{s}} \cdot V2$$

$$\sigma_{\text{n_corr}} = \sigma_{\text{n_trial}} \quad \leftarrow \text{normal traction NOT changed by slip (:202)}$$

Three facts read directly off these: 1. **Dip (1) and strike (2) are identical in form** (same η_{s} , same algebra). The solver cannot prefer one over the other — there is NO per-direction instability here. 2. **Slip onset is a switch on the magnitude $|\tau|$** (the $\max(0, \cdot)$). The $1e-14$ cross-rank difference in $|\tau|$ at the threshold flips V_{abs} between 0 and >0 , so **V1 AND V2 flip across ranks together** (they share V_{abs} and $|\tau|$). The trigger corrupts the WHOLE slip vector, not one direction. 3. **$\sigma_{\text{n_corr}} = \sigma_{\text{n_trial}}$** : slip does NOT change the normal stress *inside the solve*. And friction caps only the SHEAR ($|\tau_{\text{shear}}| \leq \mu |\sigma_{\text{n}}|$); there is no cap on the normal channel (no "no-opening" limit — this is why the original blow-up reached $\sigma_{\text{n}} = -124$ GPa tension). The normal channel is the

unconstrained one.

So why is V1 (dip) the field the guard reports, and why does the oblique run blow up while zerodip doesn't? Two parts, kept honest:

- **(reporting) Why V1 is named.** The guard uses $\text{rel_diff} = |a-b| / \max(|a|, |b|)$. When one rank slips and the other locks, BOTH V1 and V2 jump to $\text{rel_diff} = 1.0$; the verify reports the first field that hit the max, and V1 is checked before V2. So “dip” is partly a **reporting artifact** — strike desynced too. (Equation-backed: same rel_diff formula.)
- **(physics, partly hypothesis) Why the blow-up needs dip.** This is NOT derivable from the local solve (which is symmetric, above). It comes from the BULK + geometry, and the zerodip job is the experiment built to test it:
 - The local channels are decoupled, but in the BULK the imposed slip velocity (`BuildImposedState, :254–280:  $v_{\text{imp}}^{\pm} = v^{\pm} \pm (1/Z)(\tau_{\text{corr}} - \tau^{\pm})$ ,  $\sigma_{\text{imp}} = \sigma_{\text{corr}}$`) radiates and propagates. On a **32°-dipping** fault, dip slip has a vertical component that the bulk elastodynamics couple into the fault-**normal** stress next step ($\sigma_{\text{n_trial}}$ changes); strike slip (motion along the surface) does not. So a dip inconsistency leaks into the **unconstrained** normal channel (fact 3) and runs away; a strike inconsistency does not feed an un-capped channel and stays bounded.
 - **Empirical result (zerodip job 7747119, MEASURED):** zeroing the dip pre-stress separates the two effects cleanly:
 - * **Blow-up is gone** — v_{max} stays 0 to $t=0.525$ s, peaks at **~6.3 m/s** ($t=0.66$ s), then *decreases* to ~ 3 . A healthy rupture, no runaway. So the dip load IS what drives the *catastrophic blow-up* (its absence $\rightarrow$ no blow-up), consistent with the dip $\leftrightarrow$ normal-coupling hypothesis (which lives in the bulk, not the flux equations; this run is its test).
 - * **But the cross-rank inconsistency is STILL THERE** — the R-101 non-fatal log shows `worst_rel` at machine precision ($\leq 1.8e-14$) through step 1500, then jumping to **0.13 (step 1600, $t=0.56$ s) $\rightarrow$ 1.0 (step 1700)**, in fields **V2 (strike velocity) and slip2 (strike slip)**. So the inconsistency diverges in STRIKE here — it is **direction-blind** (corrects my earlier “strike works”). It coincides *in time* with the **speckled slip-rate elements outside the rupture** seen in ParaView, so the WORKING HYPOTHESIS is that the speckle is issue B. Not yet proven *spatially* (`worst_rel` is the max over shared QPs; the speckle is scattered) — **the Phase-4 post-fix rerun is the confirmation:** reconcile clears the speckle $\Rightarrow$ it WAS issue B; speckle survives at `worst_rel` $\approx 0 \Rightarrow$ a co-present cause (under-resolution $L_{\text{nuc}}/h_{\text{min}} \approx 3-10 < 10$, or SSO) the reconcile won't fix.
 - * **Net:** the cross-rank inconsistency (issue B) is present for any slip direction; whether it then BLOWS UP depends on the dip $\leftrightarrow$ normal runaway. zerodip keeps it bounded (speckle $\sim O(1)$ m/s, no blow-up). Whether the OBLIQUE (dip-loaded) run blows up or *also* stays bounded after Part A is the **OPEN test** — exactly what `spatial_dyn_resDc2_XRANKdiag*.sbatch` (now non-fatal) checks. The pre-hard-gate Dc2 run (7746601) aborted at the R-101 guard $\sim t=0.455$ s, *before* the $\sim t=1.0$ s blow-up window, so we never actually observed whether it blows up.

Bottom line for the fix. Because the local solve is direction-symmetric and the trigger corrupts the whole slip vector, the fix must make the slip decision single-valued for ALL directions — which it does. We do NOT rely on strike being self-stabilizing or on resolving the bulk dip-vs-normal question; the reconcile removes the inconsistency before it can feed any channel.

B.2.5 Scope – do TPV102/104/205 have this too? And the mixed-flux connection

Q1: yes, all three share the substrate, but only LSW is at risk. The redundant per-rank shared-QP solve from $\sim 1e-14$ -different inputs exists for EVERY config at $np > 1$. Only the AMPLIFIER differs: - **TPV102 / TPV104 (rate-and-state):** the solve $V = \text{solve}(a \cdot \text{asinh}(V \cdot C) = \tau / \sigma_n)$ (Brent, `friction_solver.cpp:34`) is **smooth & monotonic** — no $\max(0, \cdot)$ switch. A $1e-14$ input $\rightarrow \sim 1e-14$ output forever, so the cross-rank inconsistency is PRESENT but pinned at the noise floor ($\text{worst_rel} \sim 1e-14 \ll 1e-10$); never diverges, invisible. **Latent but harmless.** - **TPV205 (LSW):** SAME $V = \max(0, (|\tau| - \tau_{\text{str}}) / \eta_s)$ switch as SAFS $\rightarrow$ amplifier present $\rightarrow$ **latent-at-risk**. Not triggered in standard TPV205 because its front sweeps each point in $\sim$ one step (no slow dwell at the threshold). A TPV205-class config with a slow nucleation parking a shared QP on the threshold would diverge. - **SAFS (LSW + slow gradual overstress):** switch AND slow dwell $\rightarrow$ **triggered** (confirmed: `zerodip worst_rel` $\rightarrow 1.0$).

$\rightarrow$ This is exactly why the reconcile must be **method-invariant**: the substrate is shared by all three; gating to LSW would leave rate-state latently inconsistent and would break the day a non-smooth rate-state cap is added.

Q1, mixed-flux connection (your memory is half-right). The mixed-flux fix (`MIXED_FLUX_PLAN.md`) addressed a DIFFERENT speckle cause — **Spatial Spike Oscillations**: grid-scale ringing from the ADER predictor under-dissipating high-frequency modes at $p \geq 2$ (a numerical-DISSIPATION deficit, *consistent* across ranks). Different axis from this bug: - SAFS runs at $p=1$ (fe order: 1), where the SSO deficit is weak, AND an SSO/dissipation speckle would be cross-rank *consistent* (worst_rel small). Since SAFS shows $\text{worst_rel} \rightarrow 1.0$, the speckle here is the **cross-rank** issue, not SSO. - BUT the mixed-flux work flagged the SAME cross-rank PRINCIPLE (`MIXED_FLUX_PLAN.md` R5: “shared-face dispatch must apply mixed-flux uniformly across rank seams”; suggested an abort guard for $nprocs > 1$ AND `use_substep` AND `mixed_flux` $\neq$ none). The reconcile is the general form of that principle, applied to the friction DOFData + imposed state instead of the flux mode. So it IS the same FAMILY (cross-rank seam consistency); mixed-flux just fixed a different symptom (dissipation) on a different field.

B.3 How the proposed fix works in plain language

Stop computing the shared point twice and hoping the two answers match. Instead, **pick one rank to be the boss for that point; the boss computes the friction solve; it tells the other rank the answer; both ranks use the boss’s answer.** Now there is only ONE answer, so the two ranks cannot disagree — the yes/no slip switch is decided once, by the boss, and copied. The $1e-14$ input difference is irrelevant because the non-boss never uses its own solve.

Concretely (still plain): 1. Both ranks do their friction solve as today (cheap; we don’t restructure the solver). 2. They **exchange** the results for every shared fault point (a small MPI message — only the points on rank boundaries). 3. For each point, the **non-boss overwrites its own answer with the boss’s** — both the friction state (slip rate, corrected traction, ...) **and** the “imposed stress” that gets pushed back into the bulk wave solve. (We must copy the imposed stress too, not just the

stored numbers — otherwise the bulk would re-grow a fresh $1e-14$ difference next step and we’d be back where we started.) 4. THEN both ranks assemble their side of the flux from that single shared answer.

How the boss is chosen (no negotiation round-trip). Every shared fault point appears on exactly two ranks. The reconcile exchange (the R-101 matcher, below) pairs the two records by their face-vertex key and hands each rank its PEER’s rank number. Both ranks then independently compute the SAME value:

```
boss = min(my_rank, peer_rank)
is_boss = (my_rank == boss)          // exactly one of the two ranks is true
```

The boss keeps its own answer; the non-boss ($my_rank > peer_rank$) overwrites. “Lower rank wins” is unique, deterministic, and **purely numeric** — so, unlike the $\pm$ side label that Part A had to fix, it cannot be fooled by the curvilinear geometry. (Any fixed unique key would do; lower-rank is the cheapest because the pairing already carries $peer_rank$. We MFEM_VERIFY that each shared point pairs with exactly one peer.)

What is copied — and why the imposed stress must be too. The boss sends, per point: (a) the friction state $DofData\{V1, V2, \tau1_corr, \tau2_corr, \sigma_n_corr, slip1, slip2\}$ and (b) the canonical-frame imposed state Q_imp_plus, Q_imp_minus . The imposed state is what gets pushed back into the bulk (BuildImposedState, fault_face_flux.cpp:254–280):

```
Q_imp^±[VY] = v_t1^± ± (1/Zs)(τ1_corr − τ1^±)    Q_imp^±[SXY] = τ1_corr
Q_imp^±[VZ] = v_t2^± ± (1/Zs)(τ2_corr − τ2^±)    Q_imp^±[SXZ] = τ2_corr
Q_imp^±[VX] = v_n^± ± (1/Zp)(σ_n_corr − σ_n^±)    Q_imp^±[SXX] = σ_n_corr
```

If we copied only the stored $DofData$ but each rank assembled the flux from its OWN Q_imp (built from its own $1e-14$ -different $\tau^\pm/v^\pm$), the bulk would re-grow a fresh cross-rank difference next sub-step and we’d be back at the start. Copying the boss’s Q_imp makes the flux that BOTH ranks assemble bit-identical.

Why this is the right kind of fix (general, not a patch): it does the same thing for every friction law and every slip direction — it just makes the shared point single-valued. It does not tune a threshold, special-case the dip channel, or lean on strike being lucky. (An older “owner-broadcast” doing exactly this existed and was deleted by R-701; we are restoring it in the path R-701 left uncovered, now that we’ve proven R-701’s “both ranks agree anyway” assumption is false.)

One honest consequence (needs your OK): making the non-boss adopt the boss’s value changes TPV102/104 by $\sim 1e-14$ (they currently differ across ranks by that much; after the fix they’re bit-identical across ranks). So we trade “bit-identical to the old binary” for “physically identical ($\leq 1e-13$) AND now bit-identical across ranks” — a stronger correctness property, but it does relax the strict byte-exact rule. Flagged again here.

1.4 Constraints

- **Method- and direction-invariant:** the reconcile runs for LSW, LSW_ForcedRupture, AND rate-state; no per-law / per-direction / threshold gate.
- **Regression contract (relaxed, see B.3):** TPV102/104/205 + BP5 physically-exact

($\text{worst_rel} \leq 1\text{e-}13$ vs pre-fix) AND cross-rank bit-identical (R-101 $\text{max_rel_diff} == 0$). Overrides CLAUDE.md byte-exact. **APPROVED by user 2026-05-23** — apply the reconcile to ALL friction laws (method-invariant, per REVIEW R-001); the $\sim 1\text{e-}14$ TPV/BP5 change vs the pre-fix binary is accepted because it removes a latent cross-rank inconsistency (rationale: TPV104 already has SAFS’s slow nucleation ramp and is one friction-law change away from the same desync — see B.2.5 / the TPV-survival analysis).

- **MPI-collective-safe** (R-1600 deadlock class): every rank reaches the exchange; ranks with no shared fault faces contribute empty buffers and do not hang.
- **Boss = lower MPI rank** (unique, geometry-independent; MFEM_VERIFY exactly one boss per shared point).
- **Files Requiring Extreme Care:** `wave_operator.inl/.hpp`, `fault_face_flux.cpp`.
- Do not touch the friction solvers; do not add hardcoded constants.

1.5 Phase 0: pin the seed – magnitude + source (DIAGNOSTIC; gates the fix choice)

Goal

Close the open part of the root cause (B.1.5): measure the cross-rank input difference at the diverging QP at FULL precision, and localise its source to interpolation (1) / ghost-exchange bug (2) / bulk divergence (3) — so we know whether the reconcile (Phase 2) is the right fix or a cheaper source-fix exists.

Files to Modify

- `dynamic/wave_operator.inl` — extend the existing SEAS_DIAG_XRANK trace (ComputeADERSharedFaceFluxRHS already committed a9bd4d2/a28a5c5):
 1. Print `I_self_can` / `I_nbr_can` at **%.17e** (full double precision), not **%.10e**, so the sub-11-digit difference is visible.
 2. Add a **raw-DOF** comparison: alongside the interpolated `I*_can`, also dump the owning element’s stored DOFs that feed the QP (before `shape×DOF` interpolation) for both the local element and the ghost copy. This separates “DOFs identical but interpolation differs” (source 1) from “ghost DOFs differ from the owner’s” (source 2/3).

Detailed Requirements

1. At the target QP, print per rank, per sub-step: `I_self_can[SXX/SXY/SXZ]` and `I_nbr_can[SXX/SXY/SXZ]` at **%.17e**.
2. Print the raw face-neighbour DOF buffer entries the QP interpolates from (`nbr_data[c][nbr_idx*ndof_per_e + i]`) and the local DOFs (`Q_data[c*ndof_total_ + dof_offset1 + i]`) at **%.17e**, for one matched (local-on-A == ghost-on-B) element, so the two ranks’ copies can be diffed.
3. Run `spatial_dyn_resDc2_XRANKdiag_8N_400r_dev_2hr_safs.sbatch` (the OBLIQUE Dc2 sim, trace on, auto-extract per rank). It is now wired **non-fatal** (`SEAS_R101_NONFATAL=1 + SEAS_DIAG_BLOWUP=1`) so it does double duty:
 - (a) the **%.17e + RAWDOF** trace localises the seed (this phase); and
 - (b) dropping the hard gate lets the dip-loaded run proceed PAST the divergence, so we also see whether it stays bounded like zerodip or blows up (the B.2 open test) and it **reproduces the speckle as the pre-fix baseline** for the Phase-4 final check.

Decision (recorded in the diagnosis doc, drives Phase 2 vs a source-fix)

- **Raw ghost DOFs == owner DOFs (bit-exact) but I_{*} can differ** → source (1) interpolation → inherent → proceed to Phase 2 (reconcile).
- **Raw ghost DOFs differ from owner DOFs** → source (2) ghost-exchange bug → STOP and write a separate plan to fix the exchange (cheaper; would let R-701's no-broadcast design stand). Do NOT build the reconcile yet.
- **DOFs identical at step 0 but drift apart over steps** → source (3) bulk divergence → inherent → proceed to Phase 2.

Acceptance Criteria

- ☐ The %.17e trace shows the exact magnitude of the cross-rank input gap at the onset QP.
- ☐ The raw-DOF comparison localises the source to (1), (2), or (3).
- ☐ The diagnosis doc records the source and the resulting fix decision.

Dependencies

- Depends on: nothing (extends the shipped diagnostic). Required by: the choice of Phase 2 (reconcile) vs a source-fix.

1.6 Phase 1: reproducible local test (injection-based, np=2, both laws)

Goal

A deterministic np=2 unit test that reproduces the cross-rank split by injecting a 1-ULP input difference at the slip-onset threshold and asserts the two ranks' DOFData are bit-identical — RED before Phase 2, GREEN after — for BOTH rate-state and LSW (proving the fix is method-invariant).

Files to Create

- tests/unit/test_shared_fault_reconcile_cross_rank.cpp.

Files to Modify

- Makefile — 5 entries mirroring the tilted-test wiring (SRC ~323, OBJ ~623, link target seas_test_shared_fault_reconcile_cross_rank with the tilted test's object set, compile rule ~2719, run target test-shared-fault-reconcile-cross-rank at np=2). Not in make test.
- dynamic/fault_face_flux.hpp — behind the existing `#ifdef SEAS_TEST_INTERNAL`: static `real_t s_seas_test_qplus_xz_perturb_ulp = 0.0;` consumed in `ComputeTrialTraction` to add `s · ULP · Q_plus[SXZ]` ONLY when set (compiles out in production). (Or, if it reproduces, inject by perturbing one rank's bulk Q DOF in the test — preferred; decide at implementation.)

Detailed Requirements

1. Fixture: reuse `BuildTwoTetFaultMesh()` (planar y=0, 2 tets) + the `cy<0→rank0/cy≥0→rank1` partition from `test_rupture_multistep_serial_vs_parallel.cpp`. One shared fault face, 3 QPs.
2. Drive $|\tau|$ across $\mu_s \cdot \sigma_n$ during the run (derive load from `TPV205Params / TPV102Params`); assert at least one step has $0 < V_{abs} < V_{small}$ on one rank (the onset is reached), else FAIL “did not cross slip-onset threshold.”
3. On rank 0 only, perturb its `Q_plus[SXZ]` by exactly 1 ULP (`std::nextafter`) vs rank 1's matching input — the deterministic stand-in for the 1e-14 interpolation seed.

4. Run `kNSteps` ADER-2 steps through the threshold.
5. Each step gather both ranks' shared-QP `D0FData` via `VerifySharedFaultD0FDataConsistency(tol,&wr,&wf,/*`
(added in `a9bd4d2`); assert `wr == 0`.
6. Loop over `{RateState, LSW}` in one executable; the assertion is per-law.

Acceptance Criteria

- ☐ Builds + runs at `np=2`.
- ☐ Before Phase 2: RED — LSW `wr→0(1)`; rate-state `wr~1e-14>0` (shows the defect is method-invariant, just sub-threshold for rate-state).
- ☐ After Phase 2: GREEN — `wr == 0` for BOTH laws, every step.
- ☐ Production build (`SEAS_TEST_INTERNAL` off): hook compiles out.

Dependencies

- Depends on: nothing (built regardless of Phase 0). Required by: Phase 2 (oracle), Phase 4. Note: if Phase 0 finds a ghost-exchange bug (source 2), the oracle's GREEN target becomes the source-fix, not the reconcile — the test (cross-rank bit-identity at the threshold) is the right oracle either way.

1.7 Phase 2: the reconcile (the fix) – GO (Phase 0 confirmed source 1, job 7747304)

Phase 0 confirmed the seed is source (1) interpolation (raw DOFs bit-identical across ranks, interpolated inputs differ at $\sim 1e-14$ — see B.1.5 RESOLVED). The source (2) “fix the exchange instead” branch does NOT apply: the ghost DOFs are already bit-exact, so the only fix is to reconcile the friction output. The one remaining gate before implementation is the **byte-exact relaxation sign-off** (see B.3 / Constraints): the reconcile makes TPV102/104 cross-rank bit-identical, a $\sim 1e-14$ change vs the pre-fix binary.

Goal

After the friction solve on a shared fault face, both ranks hold bit-identical `D0FData` and assemble from the boss's imposed stress — for ALL friction laws — so the slip/lock decision is made once and copied.

Files to Modify

- `dynamic/wave_operator.inl` `ComputeADERSHaredFaceFluxRHS` ($\sim 4671-4797$): make the shared-fault QP handling **two-pass** with a reconcile between them.
- `dynamic/wave_operator.hpp`: declare the helper + assembly-buffer struct.
- Factor the R-101 record gather/pair (`:5470+`) into a shared helper used by both the verify and the reconcile (one matcher).

Detailed Requirements

1. Helper (parallel-only):

```
int WaveOperator<MeshType>::ExchangeAndPairSharedFaultQPs(
    int npay, const std::vector<double>& local_payload,
    const std::function<void(int local_qp, const double* peer_payload,
        int peer_rank)>& cb) const;
```

Gathers one record/local-shared-QP (face-vertex-key + qp_idx + rank + payload) via the existing MPI_Allgatherv, pairs by (face-key, qp_idx), invokes cb for each paired local QP. The verify uses npay=8 + a compare callback; the reconcile uses npay = 8 + 2·NUM_STATE (8 DOFData fields + I_imp_plus_can + I_imp_minus_can) + an overwrite callback.

2. Per-QP assembly buffer:

```
struct SharedFaultQPAssembly {
    int dof_offset1, ndof, dof_idx; bool elem1_on_plus; real_t w;
    real_t can_n[3], can_t1[3], can_t2[3];
    real_t shape1[MAX_NDOF];
    real_t I_imp_plus_can[NUM_STATE], I_imp_minus_can[NUM_STATE];
};
```

3. Pass 1 — compute (no assembly): run the existing friction dispatch (LSW / LSW_ForcedRupture / rate-state) → fill fault_dof_data_[dof_idx] and I_imp_*; store the buffer; do NOT touch rhs.
4. Reconcile: payload = 8 DOFData fields + I_imp_plus_can + I_imp_minus_can. Call the helper; in the callback, if peer_rank < my_rank_ (peer is boss), overwrite this rank's fault_dof_data_[dof_idx] AND the buffer's I_imp_*_can with the peer's payload. MFEM_VERIFY every local shared QP was paired.
5. Pass 2 — assemble from the (possibly overwritten) I_imp_*_can, rotated to global via T_can (rebuilt from buffered can_*, bit-identical on both ranks), exactly as the current inline assembly.
6. No friction-law gate (method-invariant). Rate-state else branch included.
7. Collective safety: the helper uses the same MPI_Allreduce(any_shared) short-circuit + MPI_Allgatherv as the verify; empty contribution from ranks with no shared fault QPs.

Edge Cases to Handle

- Unpaired shared QP → MFEM_VERIFY fail (same policy as the verify).
- Same-elem1_on_plus-on-both-ranks edge ($\theta \approx 90^\circ$) → boss-by-rank still unique.
- Serial / np=1 → no shared faces → two-pass degenerates to the current path (byte-exact).

Acceptance Criteria

- ☐ Phase-1 test GREEN for both laws (wr == 0 every step).
- ☐ TPV102/104/205 + BP5: worst_rel ≤ 1e-13 vs pre-fix AND R-101 max_rel_diff == 0.
- ☐ Local fault/TPV suite green (fault-basis trio, shared-fault role/dof-data, interior-flux ×3, godunov identity, tpv102 locked/absorbing/pepper/ ader-smoke, multistep + tilted serial-vs-parallel, sign-flipped truth table).
- ☐ np=2,4,8 incl. a partition with some rank lacking shared fault faces — no deadlock.

Dependencies

- Depends on: Phase 1. Required by: Phase 4.

1.8 Phase 3: correctness guard (runtime, method-invariant)

Goal

Keep `VerifySharedFaultD0FDataConsistency` as the permanent runtime sentinel so any future regression aborts loudly; post-fix it must report `max_rel_diff == 0`.

Files to Modify

- `dynamic/wave_operator.inl`: refactor the verify onto the shared `ExchangeAndPairSharedFaultQPs` helper (one matcher); update the stale R-501 wording in the abort message to point at the Phase-2 reconcile.
- Keep `SEAS_R101_NONFATAL` (commit fc28454) as the diagnostic escape hatch; keep the existing gated cadence; add `SEAS_VERIFY_XRANK EVERY=N` (default unchanged) for debugging.

Detailed Requirements

1. The guard stays method-invariant (checks all shared QPs regardless of law).
2. Document that post-fix the expected `max_rel_diff` is 0; any nonzero is a real regression, not roundoff.
3. Negative test (Phase 1, with the reconcile disabled via a test switch): the guard trips — proving it still catches a real desync.

Acceptance Criteria

- ☐ Post-fix Dc2 XRANK Frontera run: guard reports `max_rel_diff == 0` through and past $t \approx 0.455$ s; no abort; run advances past $t = 1.0$ s.
- ☐ Negative test trips the guard.

Dependencies

- Depends on: Phase 2. Required by: Phase 4.

1.9 Phase 4: full regression + Frontera re-validation

Goal

End-to-end: the Dc2 mesh advances past $t = 1.0$ s (no abort, no blow-up); fix is config-agnostic.

Detailed Requirements

1. Local regression battery (Phase-2 AC) green; document unchanged pre-existing failures (adjacent-triangle pepper-bug, r101 missing-precondition, macOS MUMPS Bus error).
2. Re-run the `SEAS_DIAG_XRANK Dc2 sbatch`: at the onset QP both ranks now show identical $V1/V2/\tau^*_\text{corr}$ through $t = 0.478$ s; R-101 passes; reaches t_final .
3. Re-run the physical `D_c=1.0` baseline config (not just the inflated Dc2) to confirm the fix is not tuned to Dc2.
4. **zerodip as an issue-B oracle (job 7747119 baseline = `worst_rel` → `1.0` + visible speckle)**: re-run zerodip post-fix; `worst_rel` must stay $\leq 1e-13$ (no DIVERGED) AND the speckled slip-rate elements outside the rupture must vanish. This is the cleanest already-set-up confirmation that the reconcile fixes issue B in the strike-only (non-catastrophic) case — separate from the blow-up.
5. **oblique non-fatal as the dip-channel speckle oracle (`spatial_dyn_resDc2_XRANKdiag_*.sbatch`, now non-fatal)**: the pre-fix run reproduces the speckle WITH dip traction present (and records bounded-vs-blow-up — the B.2 open test). Post-fix rerun: `worst_rel` $\leq 1e-13$ AND

the speckle clears. If the pre-fix oblique run also stays bounded (no blow-up after Part A), this run alone is the full issue-B final check. If the speckle survives at $\text{worst_rel} \approx 0$, a co-present cause (under-resolution / SSO) is implicated, NOT the reconcile (decide next steps then).

Acceptance Criteria

- ☐ Local regression green (modulo documented pre-existing).
- ☐ Frontera Dc2 AND $D_c=1.0$: no R-101 abort, no blow-up; V_{max} peaks then decreases.
- ☐ **zerodip post-fix**: $\text{worst_rel} \leq 1e-13$ for the whole run (was $\rightarrow 1.0$ at $t \approx 0.595$ s) AND no speckle in ParaView (was scattered cyan elements). V_{max} unchanged (~ 6 m/s, still physical).
- ☐ **oblique non-fatal post-fix**: $\text{worst_rel} \leq 1e-13$ AND the speckle clears in ParaView (pre-fix baseline = this run with the gate non-fatal). The bounded-vs-blow-up outcome of the pre-fix oblique run is recorded either way (B.2 test); a surviving speckle at $\text{worst_rel} \approx 0$ flags a co-present cause.

Dependencies

- Depends on: Phase 2, Phase 3.

1.10 Testing Strategy

Q2 – can we reproduce the issue locally as a guard? Three layers:

1. **Deterministic local guard (primary)**: the Phase-1 injection test ($np=2$). The clean 2-tet fixture is bit-identical across ranks, so it CANNOT make the seed naturally — we inject the 1-ULP cross-rank difference the real run gets from interpolation/bulk drift, drive to the slip-onset threshold, and assert cross-rank bit-identity. RED before Phase 2, GREEN after, for BOTH laws. It is cheap ($np=2$, a few steps) and **MUST be added to the parallel CI target** (test-parallel-class) so a future regression of the reconcile aborts CI — this is the standing local guard. (Mark it in the Makefile as a parallel test and wire it into the test-parallel aggregate, NOT only as a standalone target like the tilted test.)
2. **Natural local reproduction (optional, heavier)**: a multi-element $np \geq 2$ LSW fixture with a slow nucleation front crossing a shared QP at the threshold (a “mini-SAFS”). It would reproduce the divergence WITHOUT injection, but the seed magnitude is mesh-dependent and it is far heavier to stand up, so it is a nice-to-have, not the guard. The injection test is the reliable guard.
3. **Production runtime guard**: the Phase-3 R-101 verify (already shipped, SEAS_R101_NONFATAL for diagnostics) — post-fix $\text{max_rel_diff}==0$; a negative test (reconcile disabled) trips it. Plus the **zerodip job as a full-physics guard** (post-fix $\text{worst_rel} \leq 1e-13$ + no speckle).

Other

- **Local oracle**: Phase-1 injection test ($np=2$, both laws), RED $\rightarrow$ GREEN — proves the reconcile absorbs the seed for any friction law.
- **Guard**: Phase-3 verify, post-fix $\text{max_rel_diff}==0$; negative test trips it.
- **Regression**: TPV102/104/205 + BP5 — capture pre-fix, assert $\leq 1e-13$ AND R-101 $\text{max_rel_diff}==0$; stash-rebuild-baseline to confirm the $\sim 1e-14$ TPV change is the reconcile (non-boss $\rightarrow$ boss), not a physics change.
- **Part-A non-regression**: the zerodip job stays clean.
- **MPI robustness**: $np=2/4/8$, incl. a no-shared-fault-on-some-ranks partition.

1.11 Risk Assessment

- **Deadlock (R-1600 class):** the new exchange must be reached by every rank; detect via $np=4/8$ smoke hangs.
- **Per-sub-step exchange cost:** correctness-first; if hot on 400r, move the helper to point-to-point over the face-nbr topology (same payload). NO kink-proximity gate (re-introduces problem-specificity).
- **Two-pass refactor indexing slip** ($dof_offset1/shape1/w$): the interior path is untouched; tilted + multistep + adjacent-triangle tests cover shared-face assembly; assert Pass-2 reproduces the pre-refactor RHS when the reconcile is a no-op.
- **Byte-exact relaxation:** the $\sim 1e-14$ TPV change overrides a CLAUDE.md non-negotiable; recorded; needs sign-off.
- **Tricky existing code:** ComputeADERSharedFaceFluxRHS (4671-4797, the R-1601 inline fallback + assembly), VerifySharedFaultDOFDataConsistency (5364+, the matcher being factored), the canonical-frame reconstruction (4695-4709, must stay bit-identical across ranks for the Pass-2 T_{can} rotation).